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SPECIES AND SPECIATION

A basic task of the systematist is to estimate and describe the diversity of species in the group or faunal/floral study. From the realist perspective, one assumes that there is some true number of species in any particular group or in a particular region at any time. We strive to approach that number, but our efforts are always reduced to estimates. How accurate an estimate might be will depend on many factors. One important factor is how closely the species concept applied by the investigator approaches species as they exist in nature. Other factors also play important roles. These include the biology of the organisms, the experience of the investigator, the amount and critical nature of previous investigations, the quantity and quality of specimens available, and the analytic techniques employed.

The word *species* is applied in three distinct ways that are of interest to phylogeneticists. First, we assert that organisms living in nature, studied through examining specimens, are organized into taxa to which a binominal, or species name, is applied. These are assertions about particular species that are hypothesized to be a naturally occurring nexus of individual organisms comprising one of the basic units of evolutionary organization. Second, we seek concepts that are associated with the properties that we think all species possess. That a particular species (*Pinus ponderosa*) is an example of a particular species concept (e.g., it is an example of an evolutionary species) is an assertion that this particular species has the properties associated with the concept. Third, we have rules that govern the forming of species names and specify where in a hierarchy of classification species taxa belong. This third application is covered in Chapter 11 when we discuss taxonomy.

Four major discussions of species follow in this chapter. First, we will discuss the ontological status of species-as-taxa. This is a largely philosophical discourse. However, we think it is an important one. To not know whether binominals refer to

kinds, sets, or individuals is to admit that we know nothing of species at all (Coleman and Wiley, 2001). Second, we will discuss species concepts, the concepts that attempt to describe the properties that particular species have that make them species. Third, we will discuss modes of speciation, with particular attention paid to how these modes may affect patterns of descent. Finally, we will present some recommendations for particular decisions concerning whether a particular group of specimens should or should not be considered distinct species.

WHAT IS IT TO BE A SPECIES?

There are three basic ways of thinking about the nature of particular species: as kinds, as sets, and as individuals. As it turns out, the question has practical importance. If species are kinds, then empirical tests applied to the question of whether a group of specimens can be taken to represent a species take a different form than if species are sets and yet another form if species are individuals. As pointed out by Coleman and Wiley (2001), these questions are different from the question of whether species are real or nominal. We take the realist perspective: species exist in nature apart from our ability to perceive them and at any one time there are only so many species, not more or less depending on fancy. However, one could be a nominalist and still be faced with whether nominal species are kinds, set, or individuals.

Species as Kinds

Building on his original thesis (Ghiselin, 1966) that species are individuals, Ghiselin (1974) points out that philosophers, ever since Aristotle, have treated species as kinds/classes, while biologists, ever since Buffon, have treated species as particulars/individuals. To Ghiselin, this represents a disjunction between the philosophy of science and the practice of science. In support of his argument, Ghiselin cites (then) recent philosophers who treat species as classes: Greg (1950), Buck and Hull (1966), Lehman (1967), and Ruse (1969, 1971). Some philosophers continue to reject the thesis, searching for kind concepts that might fit. For example, Boyd (1991) suggested that species were “homeostatic cluster kinds,” a concept supported by Griffiths (1999) and Ereshefsky (2001). Others have searched for concepts of kinds that might fit species (Ruse, 1987; Mahner and Bunge, 1997; Boyd, 1999; Wilson, 1999a).

Without changing the traditional Western philosophical concept of kind, we can think of kinds as concepts associated with defining properties such that individuals (particulars) are either members of one kind or another. These defining properties are both necessary and sufficient. If a particular kind “falls out” of a scientific theory (Quine, 1969) because the theory posits that certain entities should have certain properties if the theory is true, then it is termed a *natural kind* and the properties are said to be predicted by the theory. To put it another way, scientific theories address particulars/entities through natural kinds, or kinds that are thought at the time the theory is formulated to be natural kinds such that particular examples can be found in the real world. For example, theories about population genetics contain many kinds; “Mendelian gene” and “Mendelian population” are examples. If one

can find examples of, say, a Mendelian population, then this gives a clue that population genetics theory is on the right track. However, if examples of the kind “Mendelian population” are repeatedly searched for and not found, then one would doubt that the theory of population genetics had anything interesting to say about the real world. Likewise, if evolutionary theory, broadly conceived, posits the existence of species in general, then this suggests that there is at least one natural kind, “species,” that has properties that are manifested by particular species as part of their “speciesness.” Never mind that we do not fully understand the true nature and properties of all species (or all atoms or all planets), but it would be difficult to see how we could find a single particular species without some notion of what it is to be a species that is gained from theories about the world. And we should always keep in mind the fact that if our theories about the world are wrong, then the kinds of species we think are present in the world may be an illusion.

Natural kinds, as kinds associated with general theories about processes that occur in the real world, are eternal and immutable, unbounded by either time or space, at least so long as the theory behind the kinds is considered valid. If the theory is proven false, then the kinds associated with the theory may be false as well. For example, the natural kind “helium” falls out of, or is integral to, theories of atomic physics as the natural kind of atom that has two protons. Theories of atomic physics explain why particular atoms have two protons, how having two protons confers properties such as chemical reactivity (or lack thereof), and under what circumstances one expects individual helium atoms to originate in nature.

In contrast to natural kinds, nominal kinds are simply nouns with definitions. The definitions contain certain properties, and when a thing fits the definition, the name is applied. For example, the noun *bicycle* refers to a certain kind of two-wheeled vehicle. Motorcycle is another kind of two-wheeled vehicle and automobile is a kind of four-wheeled vehicle. Motorcycle and bicycle belong to the kind “two-wheeled vehicle” and automobiles and motorcycles belong to the kind “vehicle with a motor.” No one pretends that such kinds are an integral part of some scientific theory, although all might agree that they are descriptive nouns with meaning. So the difference between natural kinds and nominal kinds is not the difference between sense and nonsense, but how such kinds relate to, or are irrelevant for, scientific theories of the world.

Wiley (1989) stressed that process theories are tested by seeing if the natural kinds they predict are fulfilled by particular entities behaving in the way predicted by the properties of the kinds. For example, atomic theory predicted that after the decay of a particular uranium atom, the result is an atom of lead and two atoms of helium. Failure to observe such a result under appropriate conditions would cause the entire theory to be suspect because predictions of the theory would not be met in nature by observing the entities (lead and helium atoms). Indeed, entire scientific theories can be overturned or extensively modified by finding one critical instance of the failure of the theory to meet the circumstances observed in nature, so long as scientists can agree that the test is valid. This, in turn, may lead to disposing of kinds once thought to be natural kinds and their replacement with kinds that are canonical to the new theory. So, natural kinds are extremely important to science as well as philosophy.

So, the question is: are particular species natural kinds? Ghiselin (2002 and earlier works from 1966) and others (e.g., Hull, 1978) argue that particular species (e.g.,

Homo sapiens and *Pinus ponderosa*) are not natural kinds at all. Species are mutable and have particular places in the fabric of history. If they have necessary and sufficient definitions, these definitions are not associated with process theories but with their time and place of origin, and in the post-Darwinian world, their ancestry. Particular species may be diagnosed, but not defined, by their character properties (Ghiselin, 1984; Wiley, 1989). Coleman and Wiley (2001) examined the issue from an analytical point of view and rejected the thesis of species-as-kinds by pointing out, among other things, that the relationship between nested kinds with transient relationships is not the relationship between particular species and the kinds with which they are associated. The line of reasoning used by them warrants some discussion.

Consider the kind “noble gas.” This kind is predicted from atomic physics as the natural kind whose member atoms have the necessary and sufficient property of having their orbitals filled with electrons. Thus these atoms are relatively unreactive (cores of old stars being an exception). Members would include *individual* atoms of helium and neon. Nested within the natural kind “noble gas” are other natural kinds, the kinds “helium,” “argon,” “neon,” and so on. The relationships of nested kinds are what we term transitive: an *individual* helium atom, being a member of the natural kind “helium,” is also a member of the natural kind “noble gas.”

Now, consider species. Evolutionary theory posits that there is at least one natural kind “species,” associated with natural processes we term speciation. Taxonomists name particular species; *Homo sapiens* is an example. Coleman and Wiley ask: does the same logical relationship exist between the kind “evolutionary species” (or “biological species” or any other kind “species”) and *Homo sapiens* that exists between noble gas and helium? To preclude arguments over words, they adopted the neutral term *constituent* rather than member (implying membership in a kind) or part (implying part of a whole). The answer is that the same logical relationship does not obtain, and considering the reasoning behind this is instructive. While an individual atom with two, and only two, protons is a constituent of both the kind “helium” and the kind “noble gas,” and an individual human being is an constituent of the taxon *Homo sapiens*, an individual human being is certainly not a constituent of the kind “evolutionary species” or any other kind of species, nor is it a monophyletic group or a natural taxon. Thus, *Homo sapiens* is not acting as a natural kind nested within the kind “evolutionary species,” any other species kind, or taxon. It is, however, a part of Hominidae, just as Ed Wiley and Bruce Lieberman are parts of *Homo sapiens* and parts of Hominidae.

Species as Sets

We must begin with the recognition that sets are not kinds. Sets are defined by extension, that is, by their membership. In contrast, kinds are defined by intension, that is, by the properties of their members. Second, sets are treated by philosophers as individuals, not as classes or kinds. Systematists are drawn to set theory because it is a well-worked system of inclusive relationships that seems to fit what systematists do in their research, especially in classification. So, it would seem natural to think of individual organisms as members of a set to which a binominal was applied. However, there are serious problems with this concept. Sets with different memberships are different sets. Owing to birth and death, the set of all humans today is different from the set of all humans yesterday. If species are sets, then there would

be many species of humans; for instance, the species that exists today, the one that existed yesterday, the one that existed two minutes ago, etc.

There is also a problem with sets and evolution. Hull (1981) argued against species-as-sets, pointing out that species, but not sets, are capable of undergoing evolutionary change. However, Kitcher (1984) argued that species were sets composed of all their present, past, and future members and thus a union of a series of subsets that exist during short time slices. He also argued that under this concept, species could be understood to change in the sense that different frequencies of traits could be observed between members of different species' subsets. To him, species change their membership over time because membership of one subset could be different from membership of a later subset. Further, one species could be understood to give rise to other species, because members of one subset of an ancestral species could be parents of members of one subset of a descendant species. Coleman and Wiley (2001) point out that Kitcher's concept of change transforms customary notions of species arising, evolving, and going extinct into one that can be explained within set theory, suggesting that the very transformation precludes species from being Kitcherian sets. He fails to show that a set-theoretical notion of species changing is an improvement upon the common, but supposedly confused notion of species changing. In addition, he does not demonstrate that the set-theoretical notion of species changing is the same as the traditional notion of species changing. Finally, the thesis of species as sets needs some notion of historical sets, a concept not yet articulated. Thus, although species can perhaps be shoehorned into Kitcherian sets, more seems to be lost than gained.

Species as Individuals

As stated above, Ghiselin (1966) and Hull (1976), as well as Hennig (1966), were among the first to seriously entertain the idea that species were individuals and not kinds. Probably because systematists had treated species as individuals for over a century (Ghiselin, 1974), biologists were quick to adopt this proposition regardless of systematic approach (cf. Wiley, 1981a; Mayr, 1982). Ghiselin has written extensively on the thesis (e.g., Ghiselin, 1997, 2002, and references cited therein) and a considerable body of literature has grown up around the notion (e.g., Coleman and Wiley, 2001). Species have many of the characteristics we look for in particular entities; they have a birth through speciation and a death through extinction. They can change their parts without changing their names, just as an individual organism can turn over its cells. They can also be named and diagnosed, but never defined, and their boundaries are fuzzy. However, there is more to the thesis. The concept is important when one considers the relationship between process theory, kinds, and entities. If species are individuals, then hypotheses about species are singular statements, and historical singular statements at that. If so, then hypotheses about species and about species' relationships are tested by weight of evidence, as outlined in Chapter 1.

SPECIES CONCEPTS

Although species-as-taxa are individuals, species concepts are kind concepts (Ghiselin, 2002 and earlier works; Wiley, 1989). As kind concepts, they are defined

by intention, each concept having properties that should provide necessary and sufficient conditions for “speciesness.” There are many species concepts, most of them reviewed by Mayden and Wood (1995) and Mayden (1997). Entire books are devoted to the subject (e.g., Mayr, 1957, 1963, 1970; Vrba, 1985; Claridge et al., 1997; Wheeler and Meier, 2000) as well as papers too numerous to cite. As kinds, we might expect that some species concepts are candidates for being natural kinds. Such concepts would “fall out” of evolutionary process theories (Quine, 1969). We might also expect some concepts to be nominal kinds, kinds that do not have direct connections with evolutionary theory but which are thought to be useful in some manner by those who invented them. Both Mayden (1997) and Wiley (2002) have suggested this distinction, and we will explore this possibility as we review some of the more prominent concepts. The goal is to see if a species concept can be found that is a natural kind concept and one that is suited for general application in phylogenetics.

Ghiselin (1974:539) began the process of sorting thorough species concepts under the paradigm of species-as-individuals. He concluded that concepts such as the Morphological Species Concept or the Phenetic Species Concept treat individual species (e.g., *Pinus ponderosa*) as “classes defined in terms of the traits of organisms rather than as individuals having the properties necessary and sufficient for membership in the species category.”

Wiley and Mayden (2000a–c) suggest that systematists form species concepts in a manner that reflects their ideas of how these concepts function in systematic and evolutionary theory. Some systematists form concepts that allow them to discover what they think are species. Others form concepts based on how they think species function in the evolutionary process. We suggest that this is nothing but the familiar debate on operationalism that surfaces from time to time in science (Wiley, 2002). Some systematists think that it is important to form “operational” concepts. Now, much depends on what one means by “operational.” If what is meant is a concept that leads to testable consequences, then, all science should be operational. However, if what is meant is the philosophy of operationalism (Bridgman, 1927), the approach is problematic. In the philosophy of operationalism, kinds and concepts are not drawn from higher level “covering laws,” but rather, they are defined by the observations by which they are measured or applied. An operational concept *sensu* Bridgman includes a discovery criterion that permits the systematist to apply some definite criteria to the question of whether a particular sample of specimens is drawn from a species or is simply a variation within a species. This approach seems seductive; it has the aroma of hard science by carefully providing specific criteria to be applied. But as pointed out by Hull (1968), operationalism carries a heavy load. Within the operational epistemology is embedded a particular ontology. The entities or properties recognized are defined by the operation employed. So if there are two operational criteria for species recognition, then there are two kinds of species. Just as there are two kinds of operational criteria for determining weight, there are two kinds of weight (English and metric come to mind, but there are more). Further, one can hardly conceive how such kinds are a consequence of process theories because they are invented by systematists.

It is exactly this distinction that led Frost and Kluge (1994), Mayden and Wood (1995), Mayden (1997), and de Querioz (1998) to suggest that a distinction can be drawn between what might be termed “general” concepts and “operational” con-

cepts. General concepts provide an ontology from which “operational” concepts (=testable concepts) may be applied. We will return to this thesis in a later section.

Process-Based Concepts

Process-based concepts attempt to characterize species in a manner predicted by “covering laws” thought to explain processes occurring in nature. Evolutionary theory (specifically that part of evolutionary theory concerned with the origin of species) provides such a “covering law,” which posits that there are two kinds of entities one might expect to find in nature that are of primary interest to systematists, species and monophyletic groups. Given that evolution appears to result in a hierarchy of entities, the monophyletic group seems to be a natural kind that falls out of the general theory of descent with modification, coupled with the general theory of speciation. Indeed, the pattern of biological diversity, the observation that it does not form a *scala naturae*, and the observation that it is largely hierarchical, seems to argue that descent with modification is not sufficient to explain biological diversity unless it is coupled with some theory of lineage splitting. Even the counter-example of the divergent hierarchy, reticulate speciation, proves instructive. Reticulate speciation is not possible unless there is something to “reticulate.” In other words, reticulate speciation depends on the prior existence of a divergent hierarchy. Otherwise, there would be no reticulate events apart from sexual reproduction.

Because all particular examples of the kind “monophyletic group” have a beginner, a common ancestral species, this suggests that species are also a necessary kind. In other words, if “monophyletic group” is a natural kind, then that from which particular monophyletic groups are derived (“species”) might also have the status of being a natural kind (Wiley and Mayden, 2000a). The results form the basis for expecting, as Hennig (1966) suggested, that organisms have two sorts of relationships. Tokogenetic relationships are formed on the basis of reproduction and obtain among individual organisms. Phylogenetic relationships are formed by severing reproductive ties to the extent that two tokogenetic systems are formed out of what was once a single tokogenetic system (or, in the case of speciation via hybridization, two tokogenetic systems form a third through tokogenesis between the two systems). For sexual organisms, tokogenesis is nonhierarchical while phylogenesis is hierarchical (speciation via hybridization is the exception). The trick is to understand how we might discover the hierarchy. Organisms access information from previous generations during their growth and development. In other words, evolution seems to be an information-conserving process in an entropic informational environment (Brooks and Wiley, 1986), and a vast literature on inheritance and ontogeny suggest that this is the case. The history of organismal evolution is rather unusual among natural processes in that we can recover at least parts of this history through studying the characters of organisms. In other words, phylogenetic descent seems an unusual natural process as it conserves some information over large time spans, permitting us to reconstruct evolutionary relationships.

What general concept of species might we develop given these observations? We suggest that the most general concept is a concept of species-as-lineages (Simpson, 1961; Hennig, 1966; Wiley, 1978, 1981a). Species are the largest tokogenetic systems of biological organization (Frost and Kluge, 1994). In Ghiselin’s (1974:538) words,

they are “the most extensive units in the natural economy such that reproductive competition occurs among their parts.” Above this level of organization are monophyletic groups and ecological communities, and each belongs to a different “scalar hierarchy” (Eldredge and Salthe, 1984; Eldredge, 1985). Scalar hierarchies are hierarchies of levels of organization, with each higher level having emergent properties not fully explained by lower levels. The “economic hierarchy” would be one familiar to ecologists (organism, population, community, ecosystem, etc.) while the “historical hierarchy” would be one familiar to phylogeneticists (organism, population, species, monophyletic group). Below the level of organization represented by species, and within the scalar hierarchy of history, are several entities including populations and individual organisms. This general concept of species is met in the Evolutionary Species Concept (ESC) discussed below. After introducing this concept, we will briefly discuss other process-based concepts, as well as several discovery-based concepts.

The Evolutionary Species Concept

The ESC, originally proposed by Simpson (e.g., 1961) is built around the notion of a singular descent community, a concept Hennig (1966) claimed could be traced at least as far back as Buffon (1749). Although Hennig (1966) favored the Biological Species Concept (BSC) of Mayr (1963), he treated species in a fashion similar to Zimmermann (1937, 1943), as individualized lineages. It is ironic that the concept most amenable to phylogenetic systematics would have been championed by an opponent of the discipline, George Gaylord Simpson (1961). The common goal for both Hennig and Simpson was a species concept with a temporal dimension. In particular, Hennig (1966:29) asserted:

We have defined the phylogenetic relationships we are trying to present as those segments of the stream of genealogical relationships that lie between two processes of speciation. Thus, by definition phylogenetic relationships exist only between species; they arise through the process of species cleavage. The key position of the species category in the phylogenetic system corresponds to the following: the species are, in the sense of class theory, the elements of the phylogenetic system. The higher categories of this system are groupings of species according to their phylogenetic relationships.

Wiley (1978) saw the similarity of Simpson’s (1961) evolutionary species concept and Hennig’s (1966) characterization of species as basic elements in phylogenetic systematics. Others, such as Ax (1987), Kluge (1990), Funk and Brooks (1990), Frost and Hillis (1990), Brooks and McLennan (1991, 2002), Frost et al. (1992), Frost and Kluge (1994), Mayden and Wood (1995), and Mayden (1997, 2002) have integrated the concept into their approaches to the phylogenetic system. The ESC has been used in studies of speciation where biologically comparable units are necessary for analysis (Wiley, 1981a; Mayden, 1985; Wiley and Mayden, 1985; Lynch, 1989; Brooks and McLennan, 1991, 2002; Frey, 1993). The most recent definition is provided below.

An evolutionary species is an entity comprised of organisms that maintains its identity from other such entities through time and over space and which has its own independent evolutionary fate and historical tendencies (Wiley and Mayden, 2000a).

Wiley and Mayden (2000a) provided an amplification of the concept in an effort to further clarify what it is to be an evolutionary species.

1. Particular evolutionary species (*Homo sapiens*, *Fundulus nottii*) are individuals with origins, existences, and ends.
2. Evolutionary species are entities within which tokogenesis prevails (Hennig, 1966).
3. A phylogenetic tree is a cartographic device (O'Hara, 1993), or graph, that portrays the hypotheses that particular species are the result of historical processes (Frost and Hillis, 1990; Kluge, 1990) and that we have discovered them during the course of research. They are, as Hennig (1966) asserted, entities that result from various processes of speciation.
4. As such, the ESC is synonymous with the Lineage Concept of Hennig (1966, not named by him but see Figures 15 and 16), the Cohesion Concept of Templeton (1989; according to Endler, 1989), the Cladistic Concept of Ridley (1986), the Concept of Population Lineages (O'Hara, 1993; de Queiroz and Gauthier, 1994), the General Lineage Concept of de Queiroz (1998), and the Hennigian Species Concept of Meier and Willmann (2000).
5. As such, the ESC is not synonymous with the BSC of Dobzhansky (1935, 1937) and Mayr (1942, 1963), the Recognition Concept of Paterson (1984), or various forms of the Phylogenetic Species Concept of Mishler and Brandon (1987), Mishler and Theriot (2000), de Queiroz and Donoghue (1988, 1990), Nixon and Wheeler (1990), or Wheeler and Platnick (2000).
6. Sexually reproducing species may show tokogenetic cohesion patterns (reproductive ties among individuals) that do not correlate with any hierarchical relationships that might exist among their parts. This is due to gene flow that might erase, for example, a time hierarchy of colonization. Contrary to the concerns of Donoghue (1985), this is to be expected based on first principles (Hennig, 1966; Kluge, 1990).
7. Completely asexual species (where they exist) have relationships among their parts that are identical to their descent relationships and are, therefore, dissimilar to sexual species and similar to clades. However, asexual species are similar to sexual species in that tokogenetic relationships predominate. In Ghiselin's (1974) terms, reproductive competition occurs among their parts. To the extent to which such species are strictly asexual, the hierarchy of such clone vectors is driven by character change and not changes in cohesion patterns. Wiley and Mayden (2000a) make a case for the inclusion of asexual species within the ESC but acknowledge that the ontology of asexual species is different from sexual species (Hennig, 1966:82–83), just as the ontology of sexual and asexual species are not identical to that of higher taxa (Wiley, 1980).
8. The ESC has been criticized for including phrases such as “maintaining identity” and “has its own independent evolutionary fate and historical tendencies” (e.g., Mayr, 2000). Similar criticisms regarding phrases such as “unitary role” have been criticized as well by Mayr (1982) who considered them unmeasurable. However, all such phrases have ontological not epistemological meaning (Wiley and Mayden, 2000a). They refer to the individuality of species. They also have testable consequences, as discussed below.

Justifications for the ESC Wiley and Mayden (2000a–c) suggest reasons for advocating the ESC in phylogenetic research. The first is a justification for the reality of species-as-lineages. The second is a justification for the assertion that species can be viewed as independently evolving lineages. The remaining justifications are specific for application of the ESC to phylogenetic systematics.

If monophyletic groups are real and the kind “monophyletic group” is a natural kind, then lineage-species must necessarily exist. Monophyletic groups are diagnosed as groups descended from a common ancestral species that was, in retrospect, the first member of the group. If the monophyletic group has reality in nature, then the ancestral lineage from which all subsequent species in the group arose must also have objective reality in nature. Discounting spontaneous generation and special creation, the only alternatives are the idea that species do not give rise to other species (Mishler and Theriot, 2000: only populations are ancestors) or that species are nominal (which amounts to the same thing because they would lack reality).

The question is: what evidence do we have that monophyletic groups exist? Empirically, the “proof” lies in the presence of synapomorphies. Synapomorphies are the lingering effects of common ancestry, just as sunlight is a “lingering effect” of Sol’s atomic chemistry. If this is true, if synapomorphies are valid evidence that monophyletic groups exist, then we can ask another question. What is the origin of a synapomorphy? At the highest level of descent, all synapomorphies begin as autapomorphies of single lineages, evolutionary innovations that became fixed subsequent to lineage splitting. How such autapomorphies became fixed is an issue that can only rarely be addressed because we cannot examine the population dynamics of lineages that existed in the past, only their descendants. We may be able to form hypotheses of why such characters persist and we may understand how they originate in the organism through studying development, but chances are slim that we will ever be able to study why they came to characterize one lineage while their homolog remained unchanged in another lineage. In fact, we may conclude that such characters were fixed in one particular ancestor only to later find evidence that it was carried as a polymorphism through several speciation events before fixation. This uncertainty seems to be the reality of historical research. Regardless of these difficulties, if at least some evolutionary innovations are real apomorphies of some monophyletic groups, then they originated as autapomorphies in single lineages, those entities termed evolutionary species. Indeed, as asserted by Hennig (1966), such an ancestral lineage is equivalent to the monophyletic group at the time of the group’s origin. Thus a monophyletic group ranked as a family was, at the time of its origin, composed of a single evolutionary species (Hennig, 1966; Wiley 1977a, 1978).

At least some ancestral species must be independently evolving entities with their own evolutionary fate and historical tendencies because synapomorphies can be used to diagnose monophyletic groups. The very fact that we are successful in reconstructing the relationships of many groups is evidence that lineages have the capacity to evolve independently from other lineages. If the origin and spread of evolutionary novelties were not highly constrained by the independence of lineages from each other and if lineages freely exchanged heritable information, then we could not reconstruct phylogenetic relationships and the history of evolutionary descent would not appear as it appears. Again, the exception proves the rule. Horizontal (lateral) gene transfer can cause havoc in phylogenetic analyses if not recognized (Daubin et al., 2003). In cases where speciation occurs via hybridization,

we may see the opposite, a confusing pattern of synapomorphies (Funk, 1985; Smith 1992) and a struggle to reconstruct evolutionary history. Of course, even in the absence of reticulation, phylogenetic relationships are not always easy to reconstruct due to unusually slow or rapid evolutionary rates in the characters examined. For instance, when rates of speciation are high relative to rates of change, as in the case of the morphology of some African rift lake cichlids (Greenwood, 1984), or the case in some diatoms (Theriot, 1992) when certain modes of speciation prevail, we may lose the trace of hierarchy.

The ESC is the logical analog of the kind “monophyletic group.” Neither is “operational,” and this is a strength of the concept, not a weakness (Wiley and Mayden, 2000a; Wiley, 2002). As we shall see, a concept can embody testable consequences without being operational *sensu* Gilmour (1940).

Application of the ESC affords the possibility that binominals (names applied to particular lineages) are comparable (in their knowledge claims if not their validity) across all groups. All evolutionary species are theoretically comparable because they all represent the largest tokogenetic systems of the organisms included in the tree of life (Frost and Kluge, 1994). They are neither parts of larger nor smaller tokogenetic systems (although they are certainly parts of monophyletic groups). A deme of *Fundulus nottii* is probably not comparable with a deme of *Pinus ponderosa*, the family Fundulidae (killifishes) certainly is not comparable to the family Pinaecae (pine trees) except that they might be monophyletic. But *F. nottii* is comparable, in theory, to *P. ponderosa* to the extent that each represents a lineage whose origin lies in a speciation event. Thus, the general characteristics of species and phenomena associated with, or causal to, speciation can be studied with more than zero degrees of freedom. We may ask questions about species in general (the properties of the kind species as it functions in evolutionary theory); we are not restricted to asking questions only about particular species. Species as basic units of biogeography become comparable, and their relationship to Earth history can be investigated. Thus, the ESC provides a biologically meaningful concept for comparing species in phylogenetics (Hennig, 1966), coevolution (Brooks and McLennan, 2002), historical ecology (Brooks, 1985), biogeography (Brundin, 1966; Brooks, 1981), speciation (Wiley, 1981a; Wiley and Mayden, 1985; Endler, 1989; Frey, 1993), and paleontology (Eldredge, 1993; Krishtalka, 1993).

The ESC provides an ontological basis for a logically consistent relationship between species and phylogenetic trees that is comparable to the relationship provided by the concept of monophyletic group *sensu* Hennig (1966). As we shall see in Chapter 4, applying the ESC to such basic questions as the number of alternative phylogenetic trees for a particular group of species or higher taxa results in far fewer alternative hypotheses than some investigators assert. Further, some of these “alternative” hypotheses turn out to be impossible alternatives.

Variations on the ESC There are a number of process-based concepts largely or entirely synonymous with the ESC. They include the following:

The Biological Species Concept of Ghiselin (1974, 1997). Ghiselin redefined the BSC as “the most extensive units in the natural economy such that reproductive competition occurs among their parts.” Frost and Kluge (1994) considered that this definition rendered the BSC “coincidental” with the ESC “at least in their application to biparentals.”

The Cohesion Species Concept (CoSC). The cohesion species concept is the most inclusive population of individuals having the potential for phenotypic cohesion through intrinsic cohesion mechanisms (Templeton, 1989:12). As stated by Templeton (1989:4), the CoSC is closely related to the ESC, differing primarily in perspective. The CoSC emphasizes mechanisms promoting cohesion while the ESC emphasizes the “manifestation of cohesion over evolutionary time.” Endler (1989) finds the concept largely synonymous with the ESC.

The Cladistic Species Concept (ClSC) of Ridley (1986) is the ESC: “A species is then that set of organisms between two speciation events, or between one speciation event and one extinction event, or that are descended from a speciation event” (Ridley, 1986:3). Ridley attempted to capture what Hennig (1966) actually meant. He emphasized the individuality of species-as-taxa and the subordinate role other species concepts have to the ESC/ClSC. Ridley’s (1986) interpretations of the ClSC relative to the ESC does differ in some respects, specifically, the persistence of ancestral species. However, we consider these differences minor. Note that de Queiroz (1998:59) classified Ridley’s concept as a variant of the Phylogenetic Species Concept, discussed more fully below. We view Ridley’s concept as an ESC process-based concept because it explicitly defines species based on the results of process and thus attempts to capture species as they are thought to exist in nature. The ClSC is closely related to the *Internodal Species Concept* of Kornet (1993).

The General Lineage Concept (GLC) of de Queiroz (1998) is the ESC. De Queiroz (1998) was attempting to show that there was a general concept that subsumes other concepts. His thesis, like that of Mayden (1997) was that all modern concepts could be grouped into concepts that were reflections on a more general concept. The difference between de Queiroz (1998) and Mayden (1997) is that Mayden made the decision that all other concepts were reflections on the ESC while de Queiroz (1998) made the decision that a new name for the ESC was needed. We do not agree.

Process-Based Concepts Emphasizing Reproductive Isolation

The most popular process-based concept emphasizing reproductive isolation is the *Biological Species Concept (BSC)* of Dobzhansky (1937) and championed by Mayr (1942, 1963, 1969, and numerous papers summarized in Mayr, 2000): A species is a group of interbreeding natural populations that is reproductively isolated from other such groups (Mayr and Ashlock, 1991:26).

Templeton (1989) has termed the BSC the “isolation concept” based on its reliance on isolating mechanisms. The BSC captures part of the ESC (Mayden, 1997; Mayden and Wood, 1995) for those evolutionary species that have achieved reproductive isolation and that are living in sympatry. Of course, sympatry is common. The question is: what is the pattern of newly formed species? The overwhelming answer is that most newly formed species live in allopatry, not sympatry (Wallace, 1855; Mayr, 1963; Lynch, 1989; Mayden and Wood, 1995; Wiley, 2002). In fact, Ghiselin (1989) has even coined “Mayr’s Law” that states “under ordinary conditions, speciation without antecedent geographic isolation does not occur” (Ghiselin, 2002:154). Most criticisms of the BSC concern either the testability or the concept in practice. Wiley and Mayden (2000a) point out that if speciation is usually allopatric (Wiley and Mayden, 1985; Lynch, 1989; Grady and LeGrande, 1992; Chesser and Zink, 1994; Mayden and Wood, 1995; Mayden, 1997; Ghiselin, 2002; Wiley, 2002; Coyne and Orr,

2004), then speciation does not require that species achieve reproductive isolation. Presumably, many newly evolved sister species live in adjacent geographic regions and in the same ecological context as the ancestral species. They never meet each other. This situation might exist for millions of years and subsequent speciation might take place in the interim. If so, and if reproductive isolation has not been achieved (as in, for example, fishes of the genus *Xiphophorus* studied by Rosen, 1979), then are we to claim that new species can arise from entities that are not, themselves, species? It seems a high price to pay when a lineage concept such as the ESC is available. It was for this reason that Paterson (1978, 1984) developed the *Recognition Species Concept (RSC)*: A species is that most inclusive population of individual, biparental organisms that share a common fertilization system. Paterson realized that organisms in populations are recognizing other members for the purposes of interbreeding rather than trying to avoid reproducing with other organisms that they could never encounter. The RSC emphasizes how isolating mechanisms keep individual species together (Templeton, 1989; de Queiroz, 1998). The concept is built around a specific mate-recognition system (SMRS) that consists of specific coadapted signals and releasing factors that are exchanged between partners and that are maintained by stabilizing selection. Such factors might operate from the level of behavior to the level of genetic recognition on the cellular level. Lieberman (1992) discussed how to potentially extend the RSC and SMRS into a phylogenetic context.

Mayden and Wood (1995) summarize other criticisms of the BSC, two of which are critical for our purposes. The BSC lacks a lineage perspective; it is nondimensional. It is difficult to translate the concept into a concept suitable for phylogenetics or paleontology, research programs that inherently contain a time dimension. Further, the BSC (and its variants) does not embrace asexual species, which are left in limbo as pseudospecies (Dobzhansky, 1970).

The *Hennigian Species Concept (HSP)* of Willmann (1986) and Meier and Willmann (2000) is difficult to characterize because it seems a mix of the BSC and the ESC. It is like an extreme form of the BSC in that complete genetic isolation relative to other species is required for species. It is like the ESC in that species should be delimited by speciation events, given that speciation involves a transition from tokogenetic to phylogenetic relationships, and the idea that phyletic speciation is an artifact and not a process (Wiley and Mayden, 2000a). Other points raised by Meier and Willmann (2000) also focused on the persistence of ancestral species. Wiley and Mayden (2000b) diverged with them on this issue, considering the persistence of ancestral species to be an open question, more in the realm of ontology than epistemology. Further, they argued that there were empirical hurdles that must be overcome before such issues could be addressed in the real world (Wiley and Mayden, 2000b, c). In short, persistence of ancestral species should, in theory, result in unresolved polytomies of descendants. The extent to which unresolved polytomies imply persistence of ancestral species is unknown and likely to remain unknown in most cases. Thus, the question is of little immediate empirical importance in actual practice, although it remains an interesting ontological issue. We also point out that while Frost and Kluge (1994) embrace the necessary extinction of ancestral species, they also embrace the ESC, implying that the issue of ancestral persistence can be considered quite apart from the issue of the validity of the ESC.

Wiley and Mayden (2000b) felt that the issue of total reproductive closure as a feature of the HSC was much more interesting. As ichthyologists, they knew of no recently evolved and closely related species of North American freshwater fishes that are 100 percent reproductively isolated (except through allopatry) and they knew of many cases where hybridization occurs among species that are not each other's closest relatives. They suggested that the demand that species be totally closed reproductive communities was extreme and not met in nature for many groups. Further, it would require sinking many species, thereby potentially masking hierarchies that have already been corroborated with synapomorphies. They suggested that occasional tokogenetic events (rare hybridization, for example) were not synonymous with defining the extent of tokogenetic systems. In this, they agreed with Mayr (1963, 1969, and other works) who has long acknowledged that some species recognized under the BSC may hybridize with other species recognized as valid under the same concept.

Phylogenetic Species Concepts

There are several versions of concepts known as the *phylogenetic species concept*. It is understandable why there are so many of these concepts. As phylogeneticists, we naturally want to distinguish ourselves by adopting phylogenetic species concepts. Some of these concepts are purely operational, but most are a mix of process-based and operational criteria. We list some of these concepts below.

Phylogenetic Species Concept I, Species as Monophyletic Taxa. A species is “a geographically constrained group of individuals with some unique apomorphous characters” (Rosen, 1978:176). To Donn Rosen, this was a relatively simple concept that makes intuitive sense and is easy to operationalize. There are many species that can be diagnosed by discovering autapomorphies (see also Nelson, 1989). A variant of this concept was articulated by Mishler and Donoghue (1982), Donoghue (1985), Mishler (1985), and de Queiroz and Donoghue (1988), who argued that one way of defining species is as monophyletic lineages that are diagnosed (“defined”) by apomorphy. Transferring the concept of monophyly from the traditional level of characterizing groups of species (Hennig, 1966) to the population level carries its own problems. Under this concept, all ancestral species are rendered paraphyletic because they would be characterized by plesiomorphies and would not include all of their descendants. Wiley and Mayden (2000a) posed the question: if the phylogenetic system allows for “paraphyletic species” (all ancestral species), then how can the phylogenetic system reject paraphyletic groups of species? One interesting response to this conundrum is to deny species ancestral status and to refer ancestry to the level of populations within species (Mishler and Theriot, 2000). Another possibility is to allow species to be defined in different ways (for example, as reproductive communities or evolutionary lineages), that is to represent different things (BSC or ESC), depending on the interests of the investigator and the scientific questions being addressed.

Phylogenetic Species Concept II, Species as Diagnosable Clusters. Eldredge and Cracraft (1980) and Nelson and Platnick (1981) suggested that it was not apomorphies but diagnosability that was the major criterion. Eldredge and Cracraft (1980:92) suggested that a species was “a diagnosable cluster of individuals within which there is a parental pattern of ancestry and descent, beyond which there is not, and which

exhibits a pattern of phylogenetic ancestry and descent among units of like kind.” Nelson and Platnick (1981:12) presented a similar concept; species are “simply the smallest detected samples of self-perpetuating organisms that have unique sets of characters.” Similar definitions were formulated by Cracraft (1983) and Nixon and Wheeler (1990). Wheeler and Platnick (2000:59) carried this line of reasoning to one logical conclusion:

Speciation is marked by character transformation. In turn, character transformation occurs through the “extinction” of ancestral polymorphism (see Nixon and Wheeler, 1992). The moment of speciation is, in theory, precise and corresponds to the death of the last individual that maintained polymorphisms within a population.

The concept has serious problems when we consider its implications to process theories. Wiley and Mayden (2000b) point out that adopting this version of the PSC amounts to divorcing cladogenesis from species entirely. For example, unless two sister species fix their respective apomorphies at the same time they will have different times of origin. This would mean that sister groups, in general, do not necessarily have the same time of origin. This would destroy one of the comparative bases of phylogenetic systematics. Adoption of such a concept would also lead to dividing tokogenetically cohesive lineages into chronospecies if more than one apomorphy is fixed at more than one time. Our reasons for rejecting chronospecies are outlined below.

Some Additional Species Concepts

The oldest and most commonly applied species concept is the *Morphological Species Concept (MSC)* (e.g., Cronquist, 1978), a concept that has been in play at least since Aristotle. Morphological species are those that have morphological differences typical of what we think of as species; this is an entirely epistemological view of species. This can reduce to the “cynical” view (Kitcher, 1984) that species are what taxonomists say they are (the *Taxonomic Species Concept*; Blackwelder, 1967). More modern incarnations of this concept are found in the *Phenetic Species Concept (PSC)* of Sokal and Sneath (1963), Sokal and Crovello (1970), Sneath and Sokal (1973), and Sneath (1976). The PSC I and II are also related, although they provide an ontological flavor to this solely epistemologically constrained concept.

The *Ecological Species Concept* of Van Valen (1976) is closely related to the ESC in some respects in that it stresses lineage independence and defines a species as a lineage that occupies a different adaptive zone than other lineages in sympatry but is evolving independently from lineages outside its range.

The rise of modern genetics and the ability to investigate genetic variation in natural populations has led to several genetics-based concepts, most of which are closely allied to the BSC. These include the *Genetic Concordance Concept* of Avise and Ball (1990) that posits that species may be identified by the concordance of multiple independent genetic markers or the closely related idea that species form genetic clusters (Mallet, 1995) or are those groups of populations fixed for unique isolating mechanisms (Wu, 2001a, b). *Compilospecies* are species that appropriate the genes of other species via interspecific hybridization (Harlan and De Wet, 1963; Aguilar et al., 1999). No doubt there could be as many species concepts of this nature

as there are interesting genetic phenomena although they would thus be limited in their relevance and entities defined as such would offer little opportunity for scientific comparisons among equivalent entities.

SORTING THROUGH SPECIES CONCEPTS

The number of papers and books written on the “species question” is large. More recent compilations include papers in Otte and Endler (1989), Claridge et al. (1997), Howard and Berlocher (1998), Wilson (1999b), and Wheeler and Meier (2000). In a series of papers, Mayden and Woods (1995) and Mayden (1997) have catalogued some 22 species concepts and others have appeared since these publications. Mayden (1997, 1999) suggests that one of the confusing aspects of the “species debate” has been the notion that the 22+ concepts are somehow equivalent in their ontological and epistemological merits, and that this is not the case. Mayden outlined four criteria (first formulated in Mayden and Woods, 1995) that a primary concept must have to function as the most general concept of species (Mayden, 1999:97).

1. The concept must have a time dimension such that the concept can be applied directly to phylogenetic hypotheses.
2. The concept must view species as individuals and not kinds so that species are considered entities.
3. The concept must be unbiased “as to the type of organism, data, or sexual tendencies of the organisms” (i.e., to asexual as well as sexual organisms).
4. The concept must be nonoperational in the sense of Bridgman operationalism (defined more fully below), but it must also have testable consequences.

Criterion 1 ties the primary species concept to phylogenetics. Criterion 2 treats species as real entities existing in nature; each of these are meritorious goals in our view. Interestingly, Criterion 2 makes species analyzable. Sets are, of course, analyzable, but we have asserted that species are not sets at all. Kinds are analyzable, but only through examining entities for regularities relative to theory prediction. If species are not entities, how can we use them to test theory? Criterion 3 suggests that the primary concept should cover asexual as well as sexual species. It also suggests that no one kind of data is inherently better (or worse) than other kinds of data that might be applied to recognizing species (for an amplification see Mayden, 2002).

Criterion 4 might sound unusual and is worth a short discussion. When Mayden (1997) states “nonoperational,” he is asserting that it should not be operational in the sense of Bridgman. That is, it should not contain a prescribed set of operations. The problem with species concepts that detail a prescribed set of operations is that they allow us to find only those species that fit the operations, thus limiting ourselves to only a part of the species-level diversity that we might discover. This is quite different from requiring a primary species concept to have testable implications (Wiley and Mayden, 2000a). In other words, a species concept might be quite “nonoperational” in the sense of Bridgman, but quite testable by examining the consequences of its definition.

Mayden and Wood (1995) found that some species concepts underestimate species diversity while others might overestimate species diversity. Mayden (1997,

1999) is interested in a primary concept that maximizes our ability to discover the actual number of species that exist, or have existed, in the world. That is, both Mayden and Wood (1995) and Mayden (1997, 1999) take a realist perspective; the metaphysical position that because species are real there must only be a finite number of them over time. (Any alternative is equally metaphysical.) Mayden (1999) groups all concepts into three categories. Most species concepts have some operational criterion embedded within the definition, rendering them strongly epistemological in nature. Some of these concepts are essentialistic (Adams, 1998). Mayden finds one concept, the ESC, to be the most robust and theoretical concept of the lot. But he also sees considerable consilience between this concept and most other concepts. In this, he agrees with de Queiroz (1998) who views most species as a manifestation of a single more general concept.

However, there is at least one concept that is not consilient with the ESC, the *Chronospecies* (Mayr, 1942) or *Successional Species* (Simpson, 1961) concept (Wiley, 1981a; Mayden, 1997). A successional species is a binominal applied to one segment of a single lineage that has, in the opinion of the describer, undergone sufficient anagenetic change to be recognizable. This concept is particularly popular among paleontologists (e.g., Gingerich, 1979; but see Krishtalka, 1993, who provides a strong criticism). Some versions of PSCII allow such species (Wheeler and Platnick, 2000). Simpson (1961) saw the need to recognize successional species because he thought that there was no nonarbitrary way to subdivide the reproductive continuum of the tree of life. Wiley (1978), citing Hennig (1966), countered that the tree was naturally subdivided by speciation events, and Wiley (1981a) rejected the concept of chronospecies because it ran counter to the thesis that species were individuals (Ghiselin, 1966) and lineages, not partial lineage segments. The ESC provides a clear alternative, but we can understand Simpson's quandary if we assume that he did not have a hierarchical perspective nor an appreciation of the difference between tokogeny and phylogeny.

Hennig (1966) saw the relationship between tokogeny and phylogeny as nontransitive. Consider this: in the absence of spontaneous generation, all life is tokogenetically connected. The nontransitive nature of the difference is apparent when we consider that if we had all of the tokogenetic relationships before us, we would be able to reconstruct the tree of life, but if all we had were the species-level relationships, we could not reconstruct the tokogeny relationships (Coleman and Wiley, 2001). Phylogeny plays by the rules of reproduction, but adds a layer of natural processes on top that we label as the various processes of speciation. Recognizing speciation events is not an arbitrary exercise, even if we miss a few and get some wrong. If we take Hennig's perspective and cleave nature at its joints, we will always separate some parent from some offspring; it is the nature of a world in which species evolve rather than a world in which species are created. If, however, we subdivide a unitary lineage, we are imposing on nature; there is no speciation event, only tokogeny.

SPECIATION: MODES AND PATTERNS

There are many avenues to choose if one wishes to study speciation. In a perspective essay, Endler (1989) asked the question: "What are we trying to explain?" Evolutionary biologists are likely to be interested in intrinsic mechanisms such as